

Problem

Obtain the initial and the final values of

$$G(s) = \frac{3s^3 + 2s + 6}{s(s+1)^2(s+1.5)}$$

Step-by-step solution

Step 1 of 2

The function $G(s)$ is,

$$G(s) = \frac{3s^3 + 2s + 6}{s(s+1)^2(s+1.5)}$$

The initial value of any function is the value obtained at time $t = 0$ in time domain or at $s = \infty$ in s domain.

Use initial value theorem to find the initial value of $G(s)$, here $g(0)$ indicates the initial value of $G(s)$ in time domain.

$$g(0) = \lim_{s \rightarrow \infty} sG(s)$$

Therefore, initial value of $G(s)$ is,

$$\begin{aligned} g(0) &= \lim_{s \rightarrow \infty} sG(s) \\ &= \lim_{s \rightarrow \infty} s \frac{3s^3 + 2s + 6}{s(s+1)^2(s+1.5)} \\ &= \lim_{s \rightarrow \infty} \frac{s(3s^3 + 2s + 6)}{s(s+1)^2(s+1.5)} \\ &= \lim_{s \rightarrow \infty} \frac{s^3 \left(3 + \frac{2}{s^2} + \frac{6}{s^3} \right)}{s^3 \left(1 + \frac{1}{s} \right)^2 \left(1 + \frac{1.5}{s} \right)} \\ &= 3 \end{aligned}$$

Therefore, the initial value of the function $G(s)$ is $\boxed{3}$.

Step 2 of 2

Final value theorem is applicable to the functions which are having poles in the left half of the s plane.

Therefore, the poles of $G(s)$ are located at $s = 0, -1, -1, -1.5$. All the poles are located in the left half of the s-plane. Hence, the final value theorem is valid for $G(s)$.

The final value of any function is the value obtained at time $t = \infty$ in time domain or at $s = 0$ in s domain.

Use final value theorem to find the final value of $G(s)$, here $g(\infty)$ indicates the final value of $G(s)$ in time domain.

$$g(\infty) = \lim_{s \rightarrow 0} sG(s)$$

Therefore, initial value of $G(s)$ is,

$$\begin{aligned} g(\infty) &= \lim_{s \rightarrow 0} sG(s) \\ &= \lim_{s \rightarrow 0} s \frac{3s^3 + 2s + 6}{s(s+1)^2(s+1.5)} \\ &= \lim_{s \rightarrow 0} \frac{(3s^3 + 2s + 6)}{(s+1)^2(s+1.5)} \\ &= \frac{(3 \cdot 0^3 + 2 \cdot 0 + 6)}{(0+1)^2(0+1.5)} \\ &= \frac{(0+0+6)}{(1^2)(1.5)} \\ &= \frac{6}{1.5} \\ &= 4 \end{aligned}$$

Therefore, the final value of the function $G(s)$ is $\boxed{4}$.

Therefore, the initial and final values of $G(s)$ are $\boxed{3, 4}$.